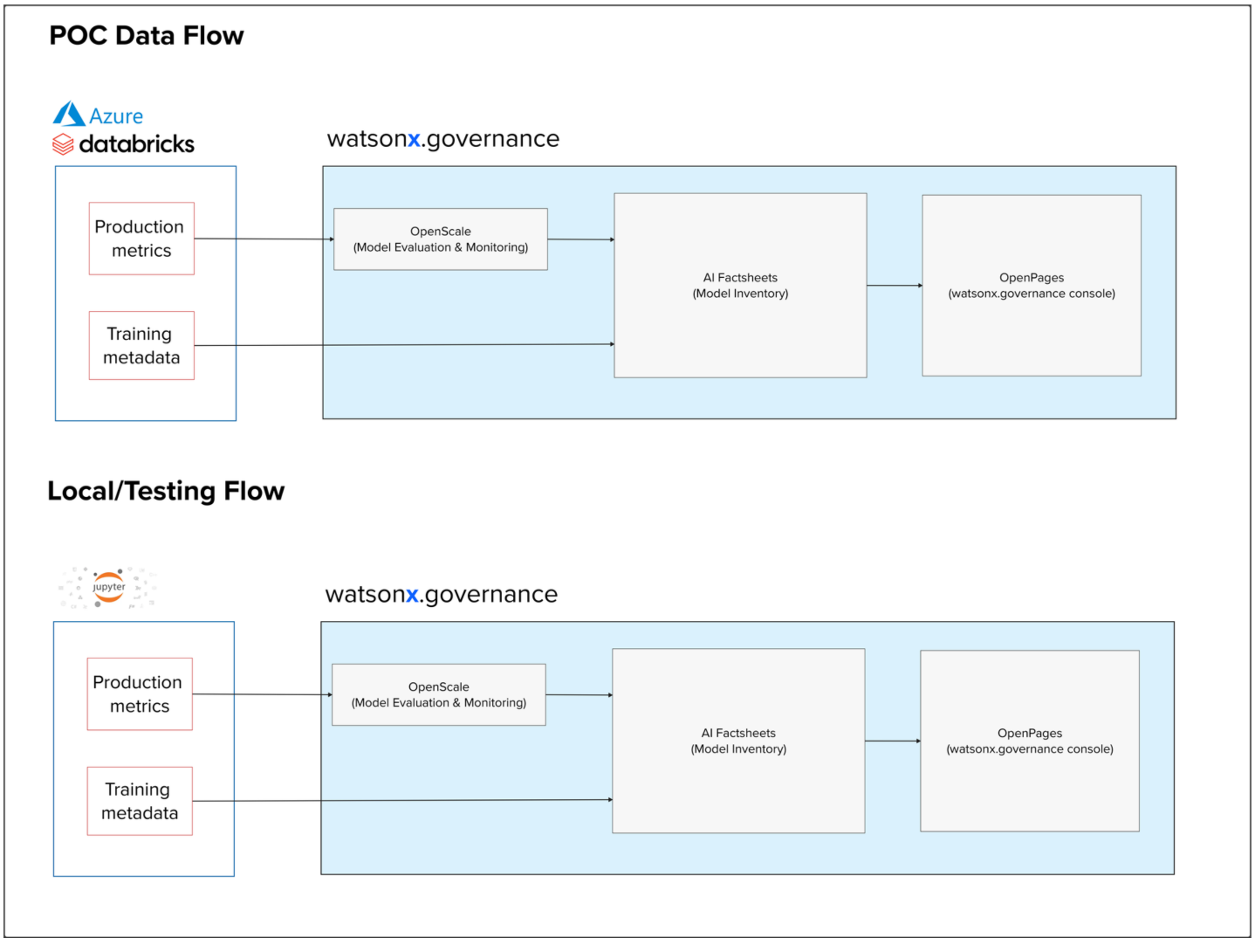
Integration with Development, Deployment and Monitoring metrics for AI Factsheets

This guide shows ideal steps to capture development and deployments facts from your development suite into AI Factsheets and linking to an approved use case, as shown below.



During the PoC development, Training and Production metrics from notebooks on Azure Databricks were utilized. However, **during the testing phase please utilize your local jupyter notebook environment to train a Machine Learning Model and capture training facts.**

Use notebook titled **Train ML Model AIFS Metrics.ipynb and** follow the below guide on important steps to follow:

- Install and import the IBM AI Gov facts client library https://pypi.org/project/ibm-aigov-facts-client/

```
In [0]:
# Package install for quickstart
!pip install ibm-aigov-facts-client
!pip install python-dotenv
```

```
In [0]:
import os
from dotenv import load_dotenv
import requests, json

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from keras.models import Sequential
from keras.layers import LSTM, Dense

from ibm_aigov_facts_client import AIGovFactsClient, CloudPakforDataConfig, DetachedPromptTemplate, PromptTemplat
```

- Initialize the facts client with your credentials for watsonx.governance

Track ML Model

Initialize fact sheets client

```
In [0]:
load_dotenv(".env")
CPD_URL = os.getenv("CPD_URL")
CPD_USERNAME = os.getenv("CPD_USERNAME")
CPD_PASSWORD = os.getenv("CPD_PASSWORD")
```

```
In [0]:
creds=CloudPakforDataConfig(
    service_url=CPD_URL,
    username=CPD_USERNAME,
    password=CPD_PASSWORD
)

# Generate a unique experiment name
# unique_suffix = str(uuid.uuid4())[:8]
experiment_name = "amazon-stock-price-0919"

# container_type= "space"
# deployment_space_id= "9815180d-01fc-4f56-87ce-dc9e907f36fd" #TD ML development Space


facts_client = AIGovFactsClient(
    cloud_pak_for_data_configs=creds,
    experiment_name=experiment_name,
    # container_type= container_type,
    # container_id= deployment_space_id,
    external_model=True,
    enable_autolog=True,
    set_as_current_experiment=True
)
```

- Train the model us for Stock price prediction using the amzn_stock_updated_v_2 dataset
- Once training is completed, create json objects for key facts you want to capture for model training as shown below

```python
training_data_reference = {
    "id": "amzn_stock_updated_v_2",
    "type": "fs",
    "location": {
        "path": "dbfs:/user/hive/warehouse/amzn_stock_updated_v_2",
        "connection": {
            "type": "dbfs"
        }
    },
    "data_type": "csv",
    "description": "Training data stored in DBFS for LSTM model"
}
model_stub_details = {
    "frameworkName": "TensorFlow",
    "frameworkVersion": "2.4.1",
    "algorithm": "LSTM",
    "epochs": 10,
    "rmse_train": rmse_train,
    "rmse_val": rmse_val
}
```

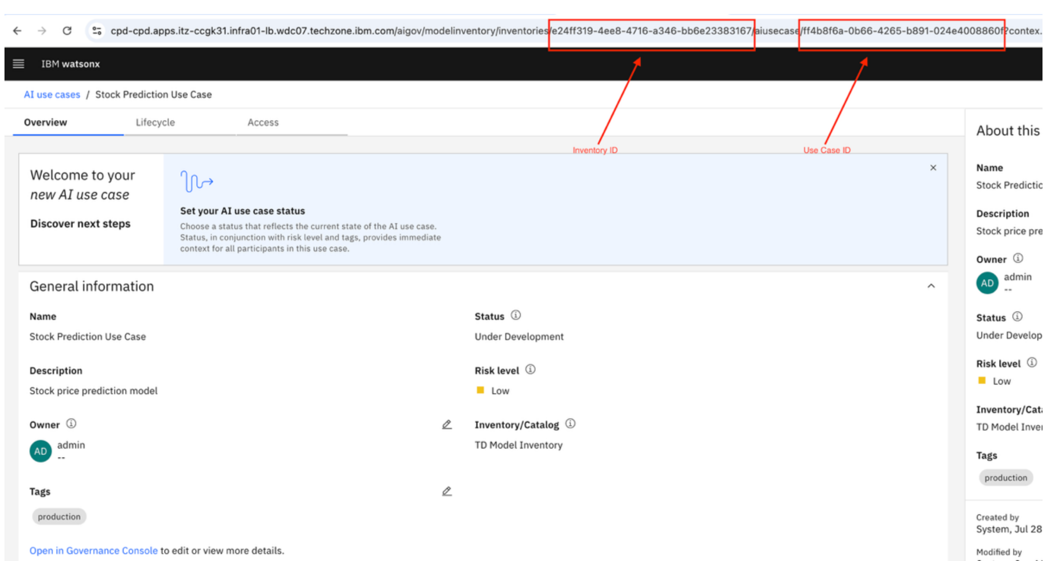
- Next export the training facts by getting the current experiment id

Export facts, save model asset

```python
current_experiment_id= facts_client.experiments.get_current_experiment_id()
facts_client.runs.list_runs_by_experiment(current_experiment_id)
```

```python
# paste in the current run ID
current_run_ID= "4b5fd743b65d4f689f666805456aae31"
facts_client.export_facts.export_payload(current_run_ID)
```

- As a final step, save the model to the inventory and track it to your use case. To get the IDs to use, open your use case in the model inventory and retrieve the IDs as shown in the screenshot below. For the approach ID, use - `"00000000-0000-0000-0000-000000000000"`



Save model to model inventory

In [0]:

```
inventory_id="e24ff319-4ee8-4716-a346-bb6e23383167" # TD Model Inventory

external_model=facts_client.external_model_facts.save_external_model_asset(model_identifier= experiment_name,
                                                                          name=experiment_name,
                                                                          model_details=model_stub_details,
                                                                          training_data_reference=training_data_
                                                                          description="Model developed in Azure
                                                                          catalog_id=inventory_id)
```

Track model to AI Use Case

In [0]:

```
usecase_id="ff4b8f6a-0b66-4265-b891-024e4008860f" # Stock Prediction Use Case
approach_id= "00000000-0000-0000-0000-000000000000" # default approach

# get AI use case
ai_usecase=facts_client.assets.get_ai_usecase(catalog_id=inventory_id,
                                              ai_usecase_id=usecase_id)
#get default approach
approach=ai_usecase.get_approach(approach_id= approach_id)

external_model.track(usecase=ai_usecase,approach=approach, version_number="0.0.8",version_comment="external model
```